Semaev first-fall replay and Boolean-parity correction

2026-10-22 — scoped validation of `crypto-autoresearcher` main `4db6ed75`

The original eight-cell summary reproduces byte for byte: 70 of 80 Semaev draws first fall at degree two, ten at degree three. Its 200 support-matched null-control grades also reproduce. The archived `fall_profile` multiplies Boolean polynomials by collecting OR-combined monomial masks in a *set*. Equal monomials should cancel over $\mathbb{F}_2$. With that parity corrected on the same frozen equations and seeds, the counts are **68 at degree two and 12 at degree three**; two nonzero-target draws change grade. Twelve of 200 null grades change. This is a source-code finding, not a new cryptanalytic timing or attack-cost measurement.

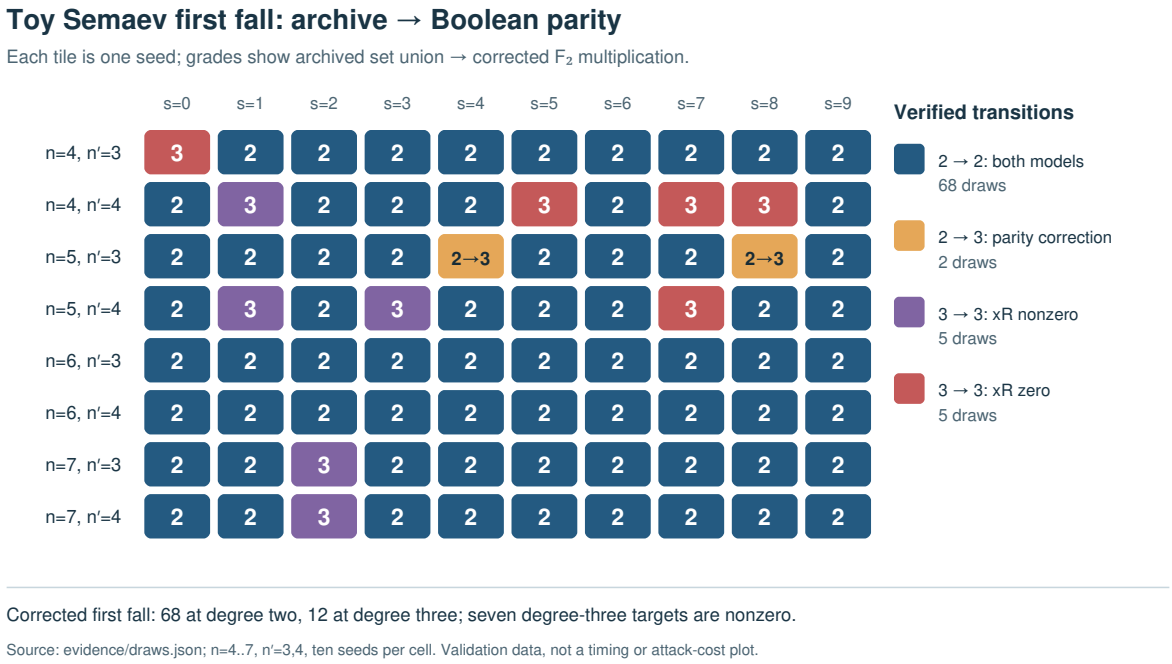


Figure: each tile shows archived → parity-corrected first-fall grade for one exact Semaev seed. Source: `evidence/draws.json`; editable renderer: `render.py`. The historical 70/10 graph remains an accurate picture of the archived table, while this chart shows the corrected Boolean result.

A direct multiplication witness

For $n = 4$, $n' = 4$, seed index $s = 1$, multiply the first component equation by u_0 (mask 1). The archived set product retains masks 1, 17, and 129; all three cancel in the $\mathbb{F}_2$ product. On the complete 256-assignment truth table, the archived product disagrees with evaluation of $f\,u_0$ in 64 assignments, and the parity-corrected product agrees in all 256. The replay checker asserts this witness, compares 21,760 finite-field products and 12,800 direct S_3 assignments, and independently computes both rank profiles on all 80 Semaev and 200 null systems. All archived distributions match the frozen table.

Finite first-fall counterexample

The same $n = 4, n' = 4, s = 1$ system has $x_R = 1$, basis $[9, 3, 2, 12]$, and field modulus $X^4 + X + 1$. Its trace residual $u_0 + u_3 + w_0 + w_3$ is already the fourth input equation. Under proper Boolean parity, $\dim V_1 = 1$ and $\dim(V_2 \cap R_{\le 1}) = 1$, hence new_falls(2) $= 0$. Also $\dim V_2 = 11$, $\dim V_3 = 53$ and $\dim(V_3 \cap R_{\le 2}) = 12$, giving new_falls(3) $= 1$. Thus first fall is three even with $x_R \neq 0$. The archived set rule instead reports $\dim V_3 = 59$ and seven new falls at three; its grade happens to agree for this draw.

Proof boundary and exact exceptions

The existing `DegreeTwoFall.lean` proves quadratic trace cancellation under $x_R \neq 0$, not a nonzero quotient class. The corrected finite example contradicts a uniform degree-two conclusion for this toy Boolean model; it does not refute the narrower Lean identity or imply a solving-degree or ECDLP cost result. A new finite Lean certificate now checks the transcribed four-equation system: its kernel-reduced ranks give no new fall at degrees one or two and one new fall at degree three. The audited theorems use only `propext`. A second Lean theorem evaluates direct $S_3 = e_2^2 + e_3 + 1$ in $\mathbb{F}_2[X]/(X^4 + X + 1)$ and the four literal component polynomials on all 256 Boolean assignments, proving equality for this exact sample. Its 256 field products and 256 direct S_3 values also match an independent Python reference. A third Lean certificate checks explicit invertible row transforms and pivots for the degree-one and degree-two matrices. Mathlib proves their $\mathbb{F}_2$ row-space dimensions are 1, 11, and 10 for degree one, degree two, and the degree-two projection; the rank difference is $11 - 10 - 1 = 0$. The executable ranks agree for this exact sample. Generic `rankOn` correctness, degree-three matrix ranks, and a symbolic S_3 descent remain open. The frozen target and historical measurement were not edited.

n	n'	seed index s	x_R	archived → corrected
4	4	1	1	$3 \to 3$
5	3	4	8	$2 \to 3$
5	3	8	8	$2 \to 3$
5	4	1	19	$3 \to 3$
5	4	3	19	$3 \to 3$
7	3	2	125	$3 \to 3$
7	4	2	113	$3 \to 3$

These seven corrected degree-three draws all have $x_R \neq 0$. Five other degree-three draws have $x_R = 0$ and lie outside the Lean theorem premise. Among the 75 nonzero-target draws, the trace residual is a new class modulo V_1 in 67 corrected degree-two cases; one other degree-two case has an old trace class but another new relation. Six degree-three cases have a nonzero trace residual already in V_1; one has a zero trace relation.

Reproduction and remaining work

The original script was run from a scratch directory and its JSON compared byte for byte with the archive (SHA-256 `64f6e34ceb16bd81…54333c99e7`). The Python 3.12 checker, `replay_ffd_semaev_20261022.py`, records every exact equation system, source hash, invocation, archived and corrected rank profile in `evidence/draws.json`; the full methodology and limits are in `report.md`. The pinned Lean source is `BooleanParityFiniteCertificate.lean`; its source hash, command and axiom output are in `evidence/lean-certificate.json` and `evidence/lean-certificate.log`. The four Lean equation lists were checked byte-for-byte against the per-draw record; see `evidence/lean-literal-check.log`. The pinned full workspace also built 2,012 jobs, including this module, and audited 475 theorem declarations with standard axioms only; see `evidence/full-workspace-replay.json`. The repository's single-target verifier returned `machine_verified` locally; its exact receipt and scratch-root hash binding are in `evidence/`. The added equation-semantics target also returned `machine_verified`; its expanded workspace built 2,013 jobs and audited 477 theorems using only standard axioms. The source hashes and exact commands are in `evidence/finite-equation-proof.json`. Hosted CI and independent semantic review remain pending. The new matrix certificate's pinned workspace built 8,959 jobs and audited 488 theorems with standard axioms only; its formal target returned `machine_verified` locally. The generator, source hashes and logs are bound in `evidence/semantic-rank-proof.json`. The first-fall chart is unchanged because no per-draw grade changed. The next formal obligation is to prove the degree-three matrix ranks and the filtration rank-nullity identity, then correct the broader frozen claim additively through review. No official research-state transition was made.